

GHOST INFRASTRUCTURE

How 19th-Century Coal Geography Still Shapes Who Gets a "15-Minute Life" Today

Executive Summary · DOI: 10.5281/zenodo.21761320 · Sakshi D. Maske

The Question

Bochum's modern shape was never designed for people — it was built around coal mines and steel works, with railways, roads, and worker-housing colonies (Zechensiedlungen) laid out to serve 19th-century industry. Coal mining ended in Bochum in 1974. More than half a century later, does that historical industrial geography still leave a measurable imprint on which neighborhoods get a genuine "15-minute city" — walkable access to essential services — and which don't?

The Method

13 historical coal mines and 4 worker-housing colonies were digitized from Mindat.org and German heritage archives. A true network-based 15-minute accessibility model was built across Bochum's complete 69,393-node pedestrian street network (OSMnx) — not a simplified straight-line radius. Historical-site proximity was statistically tested against present-day accessibility via a Welch's t-test, then verified against its most obvious confound (city-center proximity) using correlation and logistic regression, and independently corroborated with a Local Moran's I spatial-clustering analysis.

The Finding

A reversed, independently-verified effect. The hypothesis was that historical industrial sites would predict present-day neglect. The evidence showed the opposite: low-accessibility zones are, on average, further from historical industrial sites than high-accessibility zones — a highly significant, medium-to-large effect that holds independently of city-center proximity.

Metric	Value
Low-accessibility nodes — avg. distance to site	1,984 m (n = 9,858)
High-accessibility nodes — avg. distance to site	1,450 m (n = 59,535)
Welch's t-test	t = 42.887, p < 0.00001 (Cohen's d = 0.589)
Confound check — distance-to-center correlation	r = 0.063 (genuinely independent)
Logistic regression (controlling for center)	coefficient = -0.0005, p < 0.001
Local Moran's I corroboration	97.1% of low-access nodes in significant cold-spot clusters

Interpretation: 19th-century industrial cores were built dense, by necessity, around the mines and colonies where workers actually lived — that density appears to persist as present-day street connectivity and service coverage. A "path dependency of centrality," not the originally hypothesized "path dependency of neglect."

Validation & Robustness Checklist

- ✓ Full network-based accessibility model (69,393 nodes, not a straight-line radius)
- ✓ Most obvious confound explicitly tested (city-center proximity, r = 0.063)
- ✓ Logistic regression confirms independence (p < 0.001, controlling for center distance)
- ✓ Independent corroboration via Local Moran's I spatial clustering (KNN k=8, 99 permutations)
- ✓ Effect size reported, not just significance (Cohen's d = 0.589)
- ✓ Reversed-hypothesis result reported honestly, not reframed to fit the original prediction

Honest Limitation

This is a single-city case study (Bochum) — the reversed effect is well-verified within this sample but not yet tested against other post-industrial cities, so generalizability remains an open question rather than a demonstrated fact. The relationship is correlational, not causal: the confound check rules out city-center proximity specifically, but cannot rule out every possible alternative explanation. Historical mine and colony locations were digitized from secondary sources (Mindat.org, Wikipedia, regional heritage archives) rather than primary survey records.

Real-World Relevance

Urban planners often assume former industrial zones are the neighborhoods most likely to be left behind — under-served and poorly connected. This project tested that assumption directly, on real network data, and found the opposite. Getting this backwards has real planning consequences: hundreds of European cities share Bochum's 19th-century industrial-core urban form, and directing accessibility investment toward the wrong neighborhoods based on an intuitive but incorrect assumption is a real, avoidable cost.

GitHub: github.com/sakshimaske303-commits/GHOST_INFRASTRUCTURE | **Live Dashboard:**
ghostinfrastructure-areytp4x8ofu6l5tosj2z.streamlit.app | **Zenodo DOI:** 10.5281/zenodo.21761320
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